3.5 Electricity

1. Charge, Current, and Potential Difference

Electric current (I): The rate of flow of charge past a point in a circuit. Measured in amperes (A).
Equation: $I = Q / t$ or equivalently $Q = It$
Charge (Q): Measured in coulombs (C). The elementary charge (charge on one electron or proton) is $e = 1.60 \times 10^{-19}$ C.

Conventional current flows from the positive terminal to the negative terminal (historically defined before the discovery of the electron). In metallic conductors, it is actually negatively charged electrons that move in the opposite direction (from negative to positive).

Charge carriers: In metals the charge carriers are delocalised (free) electrons. In electrolytes, the charge carriers are positive and negative ions. In semiconductors, charge is carried by electrons and "holes" (vacancies where electrons are missing).

Potential difference (V): The energy transferred per unit charge passing between two points in a circuit. $V = W / Q$. Units: V (volts). $1\ V = 1\ J\ C^{-1}$.
Potential difference is sometimes called "voltage." It is always measured between two points.

Number of Charge Carriers

If N electrons pass a point in time t, the total charge is $Q = Ne$, so the current is:

$$I = Ne / t$$

The Drift Velocity Equation

In a conductor of cross-sectional area A carrying current I, with n charge carriers per unit volume each of charge q, the drift velocity v is given by:

$$I = nAqv$$

- n = number density of charge carriers (m^{-3})
- A = cross-sectional area of the conductor (m^2)
- q = charge on each carrier (C) (for electrons, $q = e = 1.60 \times 10^{-19}$ C)
- v = mean drift velocity ($m\ s^{-1}$)

In metals, n is very large (approximately 10^{28} to 10^{29} m^{-3}), so the drift velocity is very small (typically less than 1 mm s^{-1}) even for substantial currents. Semiconductors have a much smaller n, so for the same current and area, the drift velocity must be larger.

Worked Example: A copper wire has a cross-sectional area of 1.0×10^{-6} m^2 and carries a current of 3.0 A. Copper has a free-electron number density of 8.5×10^{28} m^{-3}. Calculate the drift velocity of the electrons.

Solution:
$I = nAqv$, so $v = I / (nAq)$
$v = 3.0 / (8.5 \times 10^{28} \times 1.0 \times 10^{-6} \times 1.60 \times 10^{-19})$
$v = 3.0 / (8.5 \times 10^{28} \times 1.6 \times 10^{-25})$
$v = 3.0 / 13600 = 2.2 \times 10^{-4}$ m s^{-1} (about 0.22 mm s^{-1})

This is extremely slow. The electric signal, however, travels at close to the speed of light because the electric field propagates almost instantaneously through the wire, pushing all electrons simultaneously.

Exam Tip: You might wonder how a lamp lights up instantly if drift velocity is so slow. The answer is that the electric field travels through the conductor at nearly the speed of light. When the circuit is completed, all the free electrons throughout the wire begin drifting almost simultaneously, so current flows everywhere in the circuit almost instantly.

2. Resistance

Resistance (R): The opposition to the flow of current through a component. $R = V / I$. Units: ohm (Ω). 1 Ω = 1 V A^{-1}.

Ohm's Law

> **Ohm's Law:** The current through a conductor is directly proportional to the potential difference across it, provided physical conditions (temperature, etc.) remain constant. A conductor obeying this law is called an **ohmic conductor** and has a constant resistance.

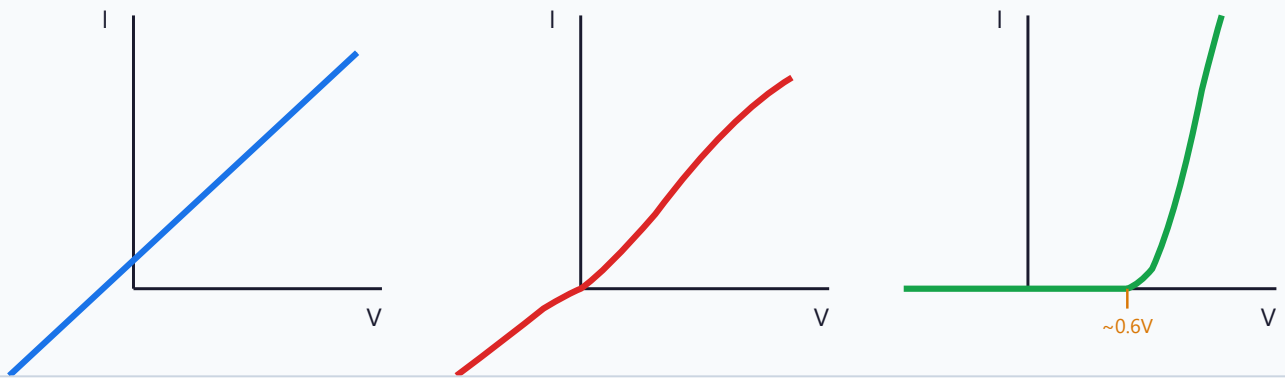
Ohm's Law is only valid when the temperature and other physical conditions are constant. A graph of I against V for an ohmic conductor is a straight line through the origin, and the resistance equals 1/gradient.

I-V Characteristics

Component	I-V Graph Shape	Description
Ohmic resistor (e.g. wire at constant temperature)	Straight line through the origin	Resistance is constant. Current is proportional to PD. Gradient = 1/R.
Filament lamp	Curved line (S-shape through origin)	Resistance increases as temperature increases (the filament gets hotter at higher currents). The gradient decreases at higher PD. The curve is symmetrical about the origin.
Semiconductor diode	Virtually zero current in reverse; exponential increase above ~0.6 V in forward bias	Very high resistance in reverse direction. Low resistance in forward direction once the threshold voltage (~0.6 V for silicon) is exceeded. Below the threshold, there is negligible current.
Thermistor (NTC)	—	Resistance **decreases** as temperature increases. At higher temperatures, more electrons gain enough energy to escape from atoms, increasing the number of charge carriers.
LDR (Light Dependent Resistor)	—	Resistance **decreases** as light intensity increases. Photons release electrons from

		atoms in the semiconductor, increasing n (number density of charge carriers).

I-V Characteristic Graphs



Microscopic Explanation of Resistance

In a metal, free electrons move through a lattice of positive metal ions. When a potential difference is applied, the electrons drift towards the positive terminal. As they travel, they collide with the vibrating lattice ions, transferring kinetic energy to them. This is the microscopic origin of resistance.

- At higher temperatures, the ions vibrate with greater amplitude, so collisions are more frequent and resistance increases.
- In an NTC thermistor, higher temperature frees more charge carriers from atoms, and this increase in n outweighs the increased collision rate, so overall resistance decreases.

Exam Tip: Be careful with thermistor types. AQA specifies NTC (Negative Temperature Coefficient) thermistors, where resistance falls as temperature rises. The physical reason is that more charge carriers are released at higher temperatures, and this effect dominates over the increased lattice vibrations. Always state the mechanism in terms of charge carriers for full marks.

3. Resistivity

$$\rho = RA / L$$

- ρ = resistivity (Ω m) — a property of the *material*, not the particular sample
- R = resistance (Ω)

- A = cross-sectional area (m^2)
- L = length (m)

Resistivity is an intrinsic property: it depends on the material and temperature, but not on the dimensions of the conductor. Rearranging gives $R = \rho L/A$, which shows that a longer wire has more resistance, and a thicker wire has less resistance.

Material	Resistivity (Ω m)	Classification
Copper	$\sim 1.7 \times 10^{-8}$	Good conductor
Constantan	$\sim 4.9 \times 10^{-7}$	Resistive alloy
Silicon	$\sim 2.3 \times 10^{3}$	Semiconductor
Glass	$\sim 10^{10}$–10^{14}	Insulator
Rubber	$\sim 10^{13}$	Insulator

Worked Example: A constantan wire has length 0.75 m and diameter 0.28 mm. Calculate its resistance. ($\rho_{constantan} = 4.9 \times 10^{-7}$ Ω m)

Solution:

First convert the diameter: $d = 0.28$ mm $= 0.28 \times 10^{-3}$ m $= 2.8 \times 10^{-4}$ m

$A = \pi(d/2)^2 = \pi \times (1.4 \times 10^{-4})^2 = \pi \times 1.96 \times 10^{-8} = 6.16 \times 10^{-8}$ m^2

$R = \rho L/A = (4.9 \times 10^{-7} \times 0.75) / (6.16 \times 10^{-8})$

$R = 3.67 \times 10^{-7} / 6.16 \times 10^{-8} =$ **6.0 Ω** (2 s.f.)

Effect of Temperature on Metals

For metals, resistivity (and therefore resistance) increases approximately linearly with temperature. As temperature rises:

1. The positive metal ions vibrate with greater amplitude.
2. Conduction electrons collide more frequently with the lattice ions.
3. The mean drift velocity decreases for a given current.
4. The number of charge carriers stays roughly constant.

Semiconductors

For semiconductors, resistivity decreases as temperature increases. Although lattice vibrations increase (as in metals), the dominant effect is the rapid increase in the number of charge carriers freed by thermal energy. The exponential increase in n outweighs the slightly reduced mobility.

Superconductors

Superconductor: A material whose resistivity drops to exactly zero when cooled below a critical temperature (also called the transition temperature, T_c). Below T_c, current flows without any energy dissipation and can persist indefinitely in a closed loop.

Applications of superconductors:

- **MRI scanners** — superconducting magnets produce strong, stable magnetic fields for medical imaging
- **Particle accelerators** (e.g. the LHC at CERN) — superconducting magnets steer and focus particle beams
- **Maglev trains** — superconducting electromagnets create levitation and propulsion
- **Power cables** — zero-resistance transmission eliminates energy losses (experimental)

Limitations: Current superconductors require cooling to very low temperatures (e.g. liquid helium at 4 K for conventional superconductors, liquid nitrogen at 77 K for some high-temperature superconductors). The cooling is expensive and energy-intensive, limiting widespread use.

Exam Tip: Be precise: a superconductor has exactly zero resistance below T_c, not "very low" resistance. This is a quantum mechanical effect (Cooper pairing of electrons) and is fundamentally different from simply cooling a normal conductor.

4. EMF and Internal Resistance

Electromotive force (EMF, ε): The total energy transferred per unit charge by the source in driving charge around a complete circuit. Units: V. *EMF is NOT a force* despite the name — it is an energy per unit charge.

Internal resistance (r): The resistance of the source itself (the battery, cell, or power supply). Energy is dissipated inside the source as thermal energy whenever current flows.

$$\varepsilon = I(R + r) = V + Ir$$

- ε = EMF of the source (V)
- I = current in the circuit (A)
- R = total external (load) resistance (Ω)
- r = internal resistance of the source (Ω)
- V = terminal PD (= IR, the PD across the external resistance)
- Ir = "lost volts" (energy wasted per unit charge inside the source)

The terminal PD (V) is always less than the EMF when current flows, because of the voltage drop across the internal resistance. When no current flows (open circuit), V = ε exactly.

Maximum Power Transfer

The current delivered to the external circuit is $I = \varepsilon/(R + r)$. The power dissipated in the external resistor is $P = I^2R = \varepsilon^2R/(R + r)^2$. This is maximised when R = r (the load resistance equals the internal resistance). At this point, exactly half the total power is delivered to the load and half is wasted internally.

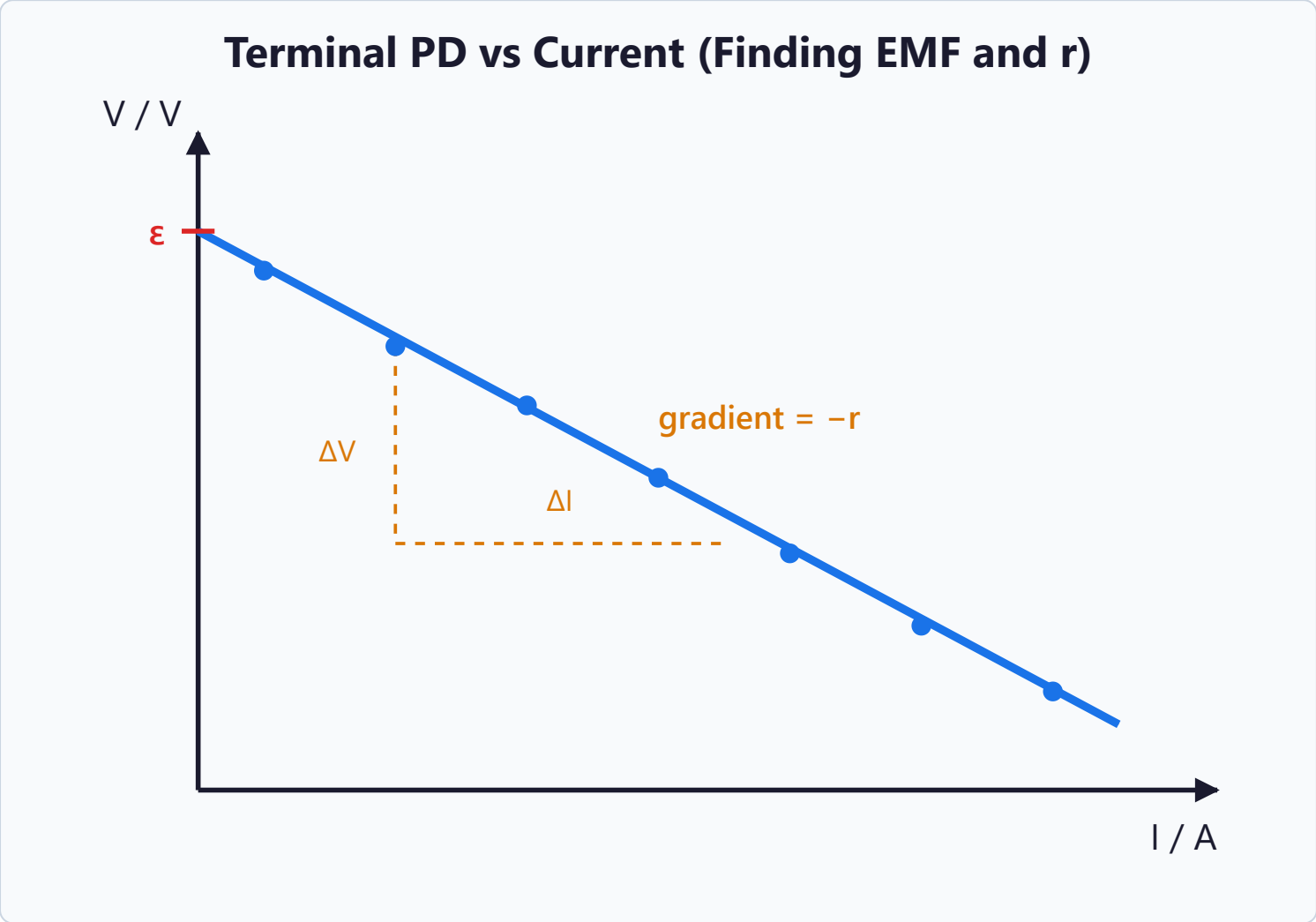
Worked Example: A battery has EMF 9.0 V and internal resistance 0.50 Ω. It is connected to a 4.0 Ω resistor. Calculate: (a) the current, (b) the terminal PD, (c) the power dissipated in the external resistor, (d) the power wasted internally.

Solution:
(a) I = ε / (R + r) = 9.0 / (4.0 + 0.50) = 9.0 / 4.5 = **2.0 A**
(b) V = IR = 2.0 × 4.0 = **8.0 V** (or: V = ε - Ir = 9.0 - 2.0 × 0.50 = 8.0 V)
(c) $P_{ext} = I^2R = 2.0^2 \times 4.0$ = **16 W**
(d) $P_{int} = I^2r = 2.0^2 \times 0.50$ = **2.0 W**
Check: total power = εI = 9.0 × 2.0 = 18 W = 16 + 2 = 18 W. Correct.

Graphical Determination of EMF and r

From V = ε − Ir, a graph of V (y-axis) against I (x-axis) gives:

- **y-intercept = ε** (this is the EMF)
- **gradient = −r** (the magnitude of the gradient is the internal resistance)
- **x-intercept = ε/r** (this would be the short-circuit current when V = 0)



Exam Tip: To find ε and r experimentally, plot V against I. The y-intercept gives ε and the gradient gives −r. A common mistake is to just measure V with a voltmeter when no current flows and assume that is ε — but even a voltmeter draws a tiny current, so graphical extrapolation is more accurate. You can also rearrange to ε = V + Ir: plot ε vs I where ε is calculated from each pair of V and I readings to check for consistency.

5. Kirchhoff's Laws

Kirchhoff's First Law (junction rule / current law): The total current entering a junction equals the total current leaving the junction. This is a consequence of *conservation of charge* — charge cannot be created or destroyed.
$\Sigma I_{in} = \Sigma I_{out}$

Kirchhoff's Second Law (loop rule / voltage law): The sum of EMFs around any closed loop in a circuit equals the sum of the PDs around that loop. This is a consequence of *conservation of energy* — the energy given to charges by EMF sources must equal the energy dissipated in components.
$\Sigma \varepsilon = \Sigma IR$

Applying Kirchhoff's Laws

To solve circuit problems:

1. Label all currents with directions (choose arbitrarily; if you get a negative answer, the current flows the opposite way).
2. Apply the junction rule at nodes where wires meet.
3. Choose closed loops and apply the loop rule. When traversing a loop, EMFs are positive if you traverse from − to + through the source, and PDs across resistors are positive in the direction of assumed current flow.
4. Solve the simultaneous equations.

Worked Example: Two cells of EMF 6.0 V and 4.0 V (both with negligible internal resistance) are connected in a loop with three resistors: 2.0 Ω, 3.0 Ω, and 5.0 Ω in series. The 6.0 V cell drives current clockwise and the 4.0 V cell opposes it. Find the current.

Solution: Applying Kirchhoff's second law around the loop (clockwise):
$\Sigma \varepsilon = 6.0 - 4.0 = 2.0$ V (the 4.0 V cell opposes, so its EMF is subtracted)
$\Sigma IR = I(2.0 + 3.0 + 5.0) = 10.0I$
$2.0 = 10.0I$
$I =$ **0.20 A** (clockwise, confirming our assumed direction)

6. Series and Parallel Circuits

Series Circuits

- Same current through all components: $I_{total} = I_1 = I_2 = ...$
- PDs add up: $V_{total} = V_1 + V_2 + ...$
- Resistances add: $R_{total} = R_1 + R_2 + ...$
- The total resistance is always greater than the largest individual resistance.

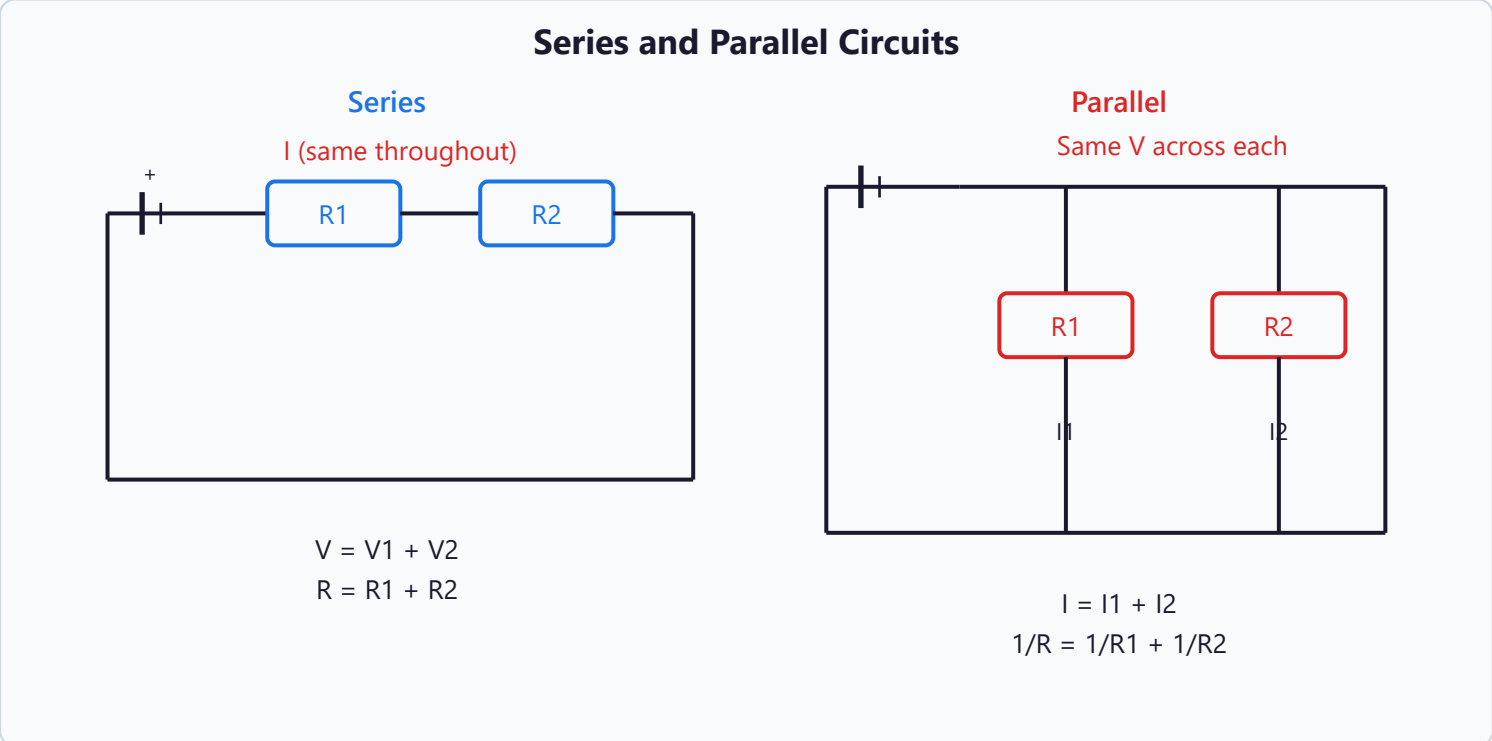
Parallel Circuits

- Same PD across each branch: $V_{total} = V_1 = V_2 = ...$
- Currents add up: $I_{total} = I_1 + I_2 + ...$
- Reciprocal of resistances add: $1/R_{total} = 1/R_1 + 1/R_2 + ...$
- The total resistance is always less than the smallest individual resistance.

Shortcut for Two Resistors in Parallel

$$R_{total} = (R_1 \times R_2) / (R_1 + R_2)$$

This "product over sum" formula only works for exactly two resistors. For more, use the reciprocal formula.



Worked Example: A 6.0 Ω resistor and a 12 Ω resistor are connected in parallel. This combination is then connected in series with a 4.0 Ω resistor. Calculate the total resistance.

Solution:

Step 1 — Parallel combination:

$R_{parallel}$ = (6.0 × 12) / (6.0 + 12) = 72 / 18 = 4.0 Ω

Step 2 — Series with 4.0 Ω:

R_{total} = 4.0 + 4.0 = **8.0 Ω**

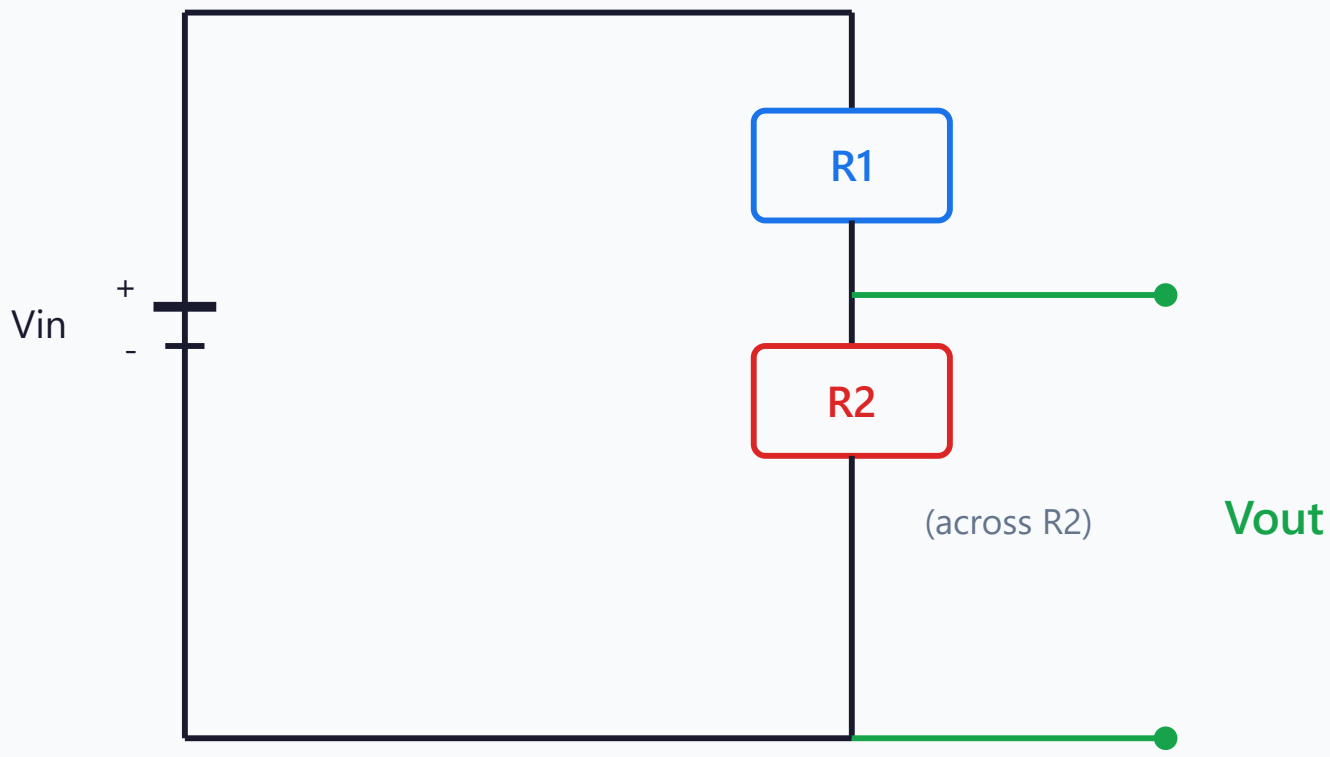
7. Potential Dividers

A potential divider consists of two (or more) resistors in series connected across a supply voltage. The output voltage is taken across one of the resistors.

$$V_{out} = V_{in} \times R_2 / (R_1 + R_2)$$

This formula comes directly from the fact that in a series circuit the PD across each resistor is proportional to its resistance (since V = IR and I is the same for both).

Potential Divider



$$V_{out} = V_{in} \times R_2 / (R_1 + R_2)$$

Sensor Circuits Using Potential Dividers

Potential dividers with variable resistors (thermistors, LDRs) can be used as sensor circuits. The key is understanding which resistor V_{out} is measured across:

Sensor	Condition Change	V_{out} across sensor	V_{out} across fixed resistor
NTC Thermistor	Temperature rises	Decreases ($R_{thermistor}$ falls)	Increases
NTC Thermistor	Temperature falls	Increases ($R_{thermistor}$ rises)	Decreases
LDR	Light increases	Decreases (R_{LDR} falls)	Increases
LDR	Light decreases	Increases (R_{LDR} rises)	Decreases

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Worked Example: An LDR (resistance 2.0 kΩ in bright light, 50 kΩ in darkness) is connected in series with a 10 kΩ fixed resistor across a 5.0 V supply. V_{out} is measured across the fixed resistor. Calculate V_{out} in bright light and in darkness.

Solution:
In bright light: R_{LDR} = 2.0 kΩ
V_{out} = 5.0 × 10 / (2.0 + 10) = 50 / 12 = **4.2 V**

In darkness: R_{LDR} = 50 kΩ
V_{out} = 5.0 × 10 / (50 + 10) = 50 / 60 = **0.83 V**

So V_{out} across the fixed resistor is high in bright light and low in darkness. This could trigger an alarm system to switch on lights when it gets dark (using a comparator circuit).

Exam Tip: Think carefully about which resistor V_{out} is measured across. If V_{out} is across the thermistor, then V_{out} decreases when temperature increases. If V_{out} is across the fixed resistor, V_{out} increases. A common mistake is to assume the output always increases with the condition — it depends on which resistor you take the output across.

Exam Tip (Loading Effect): The potential divider equation only applies when no current is drawn from the output terminals. If a load (e.g. a voltmeter or another component) is connected across V_{out}, it acts in parallel with R_2, reducing the effective resistance and therefore reducing V_{out} below the calculated value. This is called the "loading effect."

8. Electrical Power and Energy

Electrical power (P): The rate at which energy is transferred by an electrical component. Units: W (watts). 1 W = 1 J s^{-1}.

$$P = IV$$

Using V = IR, we can derive two alternative forms:

$$P = I^2R = V^2/R$$

Energy Transferred

$$E = Pt = IVt = I^2Rt = V^2t/R$$

Units: J (joules). In domestic contexts, energy is often quoted in kilowatt-hours (kWh):

$$1\ \text{kWh} = 1000\ \text{W} \times 3600\ \text{s} = 3.6 \times 10^6\ \text{J} = 3.6\ \text{MJ}$$

Worked Example: A 60 W filament lamp operates at 230 V mains. Calculate: (a) the current through the lamp, (b) the resistance of the filament when operating, (c) the energy used in 8.0 hours, in kWh and joules.

Solution:
(a) I = P / V = 60 / 230 = **0.26 A**
(b) R = V / I = 230 / 0.26 = **880 Ω** (or R = V^2/P = 230^2/60 = 880 Ω)
(c) E = Pt = 60 × 10^{-3} kW × 8.0 h = **0.48 kWh**
In joules: E = 0.48 × 3.6 × 10^6 = **1.73 × 10^6 J** (1.7 MJ)

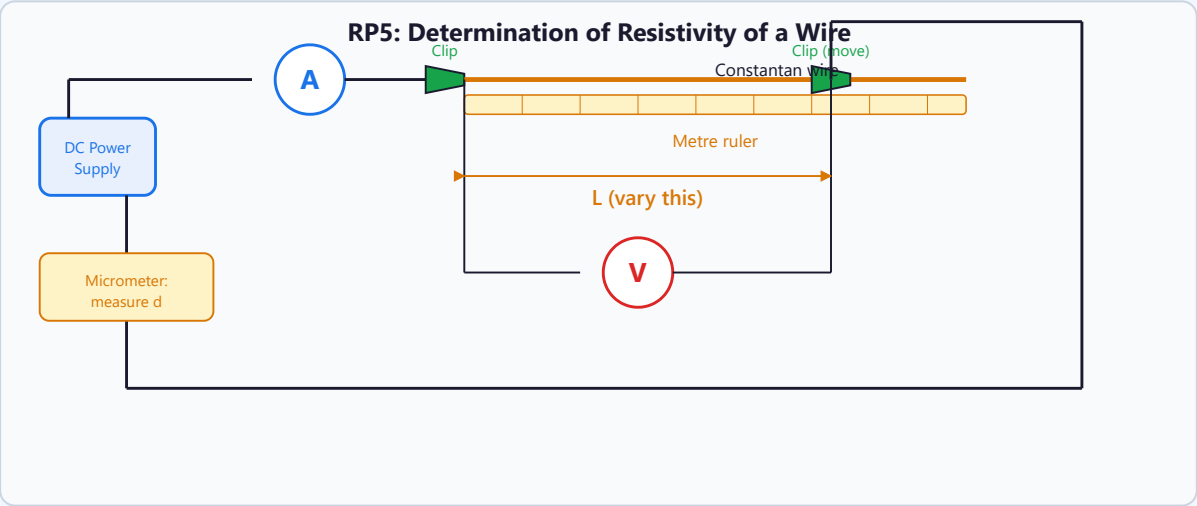
Exam Tip: Choose the power formula based on what you know. If you know I and R but not V, use P = I^2R. If you know V and R but not I, use P = V^2/R. This avoids unnecessary intermediate calculations and reduces rounding errors.

9. Summary of Key Equations

Quantity	Equation	Units
Current	I = Q/t = nAqv	A
Potential difference	V = W/Q	V
Resistance	R = V/I	Ω
Resistivity	ρ = RA/L	Ω m

EMF / internal resistance	$\varepsilon = I(R + r) = V + Ir$	V
Resistors in series	$R_T = R_1 + R_2 + \ldots$	Ω
Resistors in parallel	$1/R_T = 1/R_1 + 1/R_2 + \ldots$	Ω
Potential divider	$V_{out} = V_{in} R_2/(R_1 + R_2)$	V
Power	$P = IV = I^2R = V^2/R$	W
Energy	$E = Pt = IVt$	J

⚗ Required Practical 5: Determination of Resistivity of a Wire



Apparatus

- Constantan wire (SWG 28 or 32, taped to a metre ruler)
- Crocodile clips (one fixed, one moveable along the wire)
- DC power supply (low voltage, e.g. 1–2 V)
- Ammeter (in series)
- Voltmeter (in parallel across the wire section)
- Metre ruler
- Micrometer screw gauge (to measure wire diameter)
- Switch (to prevent heating between readings)

Method

1. Tape the constantan wire along a metre ruler. Connect crocodile clips at the ends of the section to be tested.
2. Connect the ammeter in series and the voltmeter in parallel across the wire between the clips.
3. Measure the wire diameter d with a micrometer at several points (at least 6 readings at different positions and orientations). Calculate the mean diameter and cross-sectional area $A = \pi(d/2)^2$.
4. Set the moveable clip to give a length L (e.g. start at 0.100 m).
5. Switch on, quickly read V and I, then switch off (to prevent heating). Calculate $R = V/I$.
6. Move the clip to increase L in steps (e.g. 0.100 m increments up to 1.000 m). Record V and I at each length.
7. Plot R (y-axis) against L (x-axis). This should be a straight line through the origin.
8. Gradient = ρ/A, so resistivity ρ = gradient × A.

Safety

- Use a low voltage and switch off between readings to prevent the wire heating up (which would change its resistance).
- Do not touch the wire while current is flowing — it may become hot.
- Ensure all connections are secure to avoid sparking.

Analysis

From $R = \rho L/A$, a graph of R vs L is a straight line through the origin with gradient = ρ/A.

Therefore ρ = gradient × A. The accepted value for constantan is approximately 4.9×10^{-7} Ω m.

If the line does not pass through the origin, this suggests a systematic error (e.g. contact resistance at the clips, which adds a constant offset to all R values).

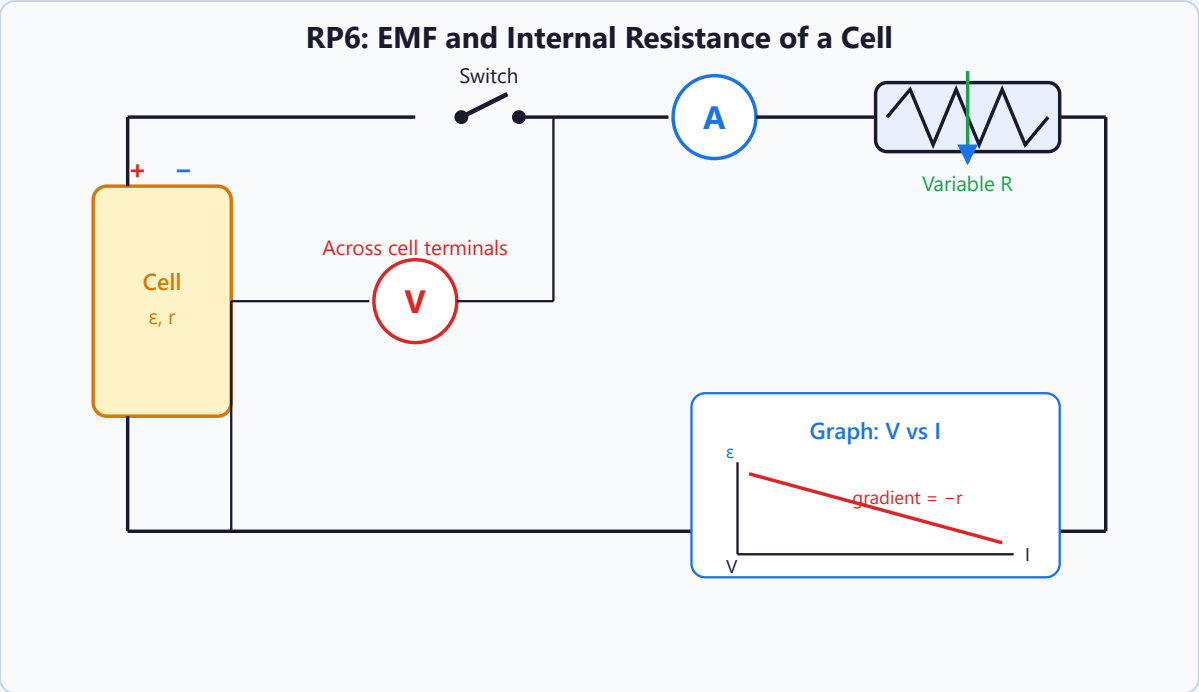
Sources of Error & Improvements

Source of Error	Type	Improvement
Wire heats up during measurement, changing resistance	Systematic	Use low current; switch off between readings; allow cooling time

Wire diameter varies along its length	Random	Measure diameter at multiple points and orientations; take a mean
Contact resistance at crocodile clips	Systematic (gives R vs L graph a y-intercept)	Ensure clips are clean and tightly attached; the y-intercept represents contact resistance
Parallax error reading the ruler for L	Random	Read the ruler from directly above; use a set square
Wire may not be perfectly straight	Systematic	Tape the wire firmly along the ruler; do not stretch it

Exam Tip: The biggest percentage uncertainty is usually in the diameter measurement. Since $A = \pi d^2/4$, the percentage uncertainty in A is **twice** the percentage uncertainty in d. Always state that you measure the diameter at multiple points and orientations and take a mean. Constantan is chosen because its resistivity barely changes with temperature, minimising heating errors.

⚗ Required Practical 6: Investigation of EMF and Internal Resistance



Apparatus

- Cell or battery (e.g. 1.5 V AA cell)
- Variable resistor (rheostat) or set of fixed resistors of different values
- Ammeter (in series)
- Voltmeter (in parallel across the cell terminals)
- Switch
- Connecting leads

Method

1. Connect the cell in series with the switch, ammeter, and variable resistor. Connect the voltmeter across the cell terminals (not across the resistor).
2. Close the switch and record the voltmeter reading (V, terminal PD) and ammeter reading (I).
3. Adjust the variable resistor to change the current. Record at least 6–8 pairs of V and I values across a wide range of currents.
4. Open the switch between readings to prevent the cell from running down and heating up.
5. Plot a graph of V (y-axis) against I (x-axis).
6. The y-intercept = EMF (ε). The gradient = −r (negative of internal resistance).

Safety

- Do not short-circuit the cell (i.e. do not set R = 0) — this draws very high current and may cause overheating.
- Use a switch and open it between readings to prevent the cell from heating up and running down.
- Handle cells carefully; do not connect in reverse or puncture them.

Analysis

From $\varepsilon = V + Ir$, rearranging: $V = \varepsilon - Ir$. This has the form $y = mx + c$:

- **y-intercept = ε** (EMF of the cell)
- **gradient = −r** (negative of internal resistance)

Typical values: $\varepsilon \approx 1.5$ V for an AA cell, $r \approx 0.5$–$1.5\ \Omega$.

Sources of Error & Improvements

Source of Error	Type	Improvement
Cell runs down during experiment (EMF decreases)	Systematic	Take readings quickly; open the switch between readings; use a fresh cell
Cell heats up at high currents, changing r	Systematic	Keep currents moderate; allow cooling between readings
Voltmeter draws a small current (reads slightly less than true EMF)	Systematic	Use a high-resistance voltmeter (digital voltmeter preferred)
Contact resistance in connections	Systematic	Use clean, tight connections; solder if possible
Limited range of I values	Random	Use a wide range of resistor values to get a good spread of data points

Exam Tip: The EMF is the y-intercept (not any single voltmeter reading). The internal resistance is the magnitude of the gradient. A steeper negative gradient means higher internal resistance. If asked how to find ε without drawing a graph, measure V when I = 0

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(open circuit) — but in practice the voltmeter draws a tiny current, so graphical extrapolation is more accurate.